

# Introduction to IOT:

## Definition:

- IOT is the system of physical devices & things that can be connected to each other through the help of internet for sharing of data.

## Characteristics (How it works)

- Sensors
- Connectivity of internet
- low power consumption
- Cloud storage
- Communication devices
- Remote access
- Security

## Advantages

- Save time
- More Safety
- Health Monitoring.
- Smart City

There are two ways of building IOT.

- 1) From a separate internetworking including only physical objects.
- 2) Make the internet ever more expansive, but this requires hard-core technologies such as rigorous cloud computing and rapid big data storage.

## # IOT advantages:

- The advantages of IOT span across every area of lifestyle and business. Here is a list of some of the advantages that IOT has to offer.
- Improved customer Engagement:
  - current analytics suffer from blind-spots and significant flaws in accuracy; and as noted, engagement remains passive.
- Technology optimization:
  - The same technologies and data which improve the customer experience also ~~known~~ to improve device use.
- Reduced waste
  - IOT makes areas of improvement clear.
- Enhanced Data Collection:
  - Modern data collection suffers from its limitations and its design for passive use.



## # IOT devices and Embedded Devices

- An embedded device is a small computer that is ~~fixed~~ used inside a machine. that can do one task. It cannot connect to the internet and work by itself.
- An IOT device is a smart device that can connect to internet, using sensors it sense the data, and sends or receives the data through internet. It can perform multiple tasks.

→

### Difference between IOT device and Embedded devices:

#### IOT devices

#### Embedded devices

- |                                                                  |                                                               |
|------------------------------------------------------------------|---------------------------------------------------------------|
| - Is a smart device that can connect to internet and share data. | - A device that do a fixed job using a small computer inside. |
| - It can connect to internet.                                    | - It cannot connect to internet.                              |
| - Smart control through the help of mobile phone.                | - Does not support any smart control.                         |
| - can get updates online                                         | - cannot get's updates                                        |

Ex: Smartwatch, smart city, motion sensors with data detection / collection

Eg: washing machines digital watches, calculators, microwave ovens.

## # Properties of IOT devices:-

- i) Connectivity
- ii) Sensors, Actuators: works on the sensed data
- iii) Processing
- iv) Cloud Storage
- v) Security
- vi) Low power consumption
- vii) Scalability

## # IOT Ecosystem:-

- An IOT Ecosystem is the complete system that includes everything to design IOT device that work together like Hardware, Software, Network, Users

### Parts of IOT Ecosystem

- i) Device / sensors: It collects data from the environment.
- ii) Network: Sends data to the cloud on application
- iii) Data processing: It process the data and convert it into useful information.
- iv) Cloud: Sends the data to the cloud to store safely.
- v) Applications: user see and control IOT devices
- vi) User: The person who uses the IOT system



## # IOT decision framework:-

- An IOT decision framework is a step by step guide to help people to plan and build IOT systems by making smart choices.

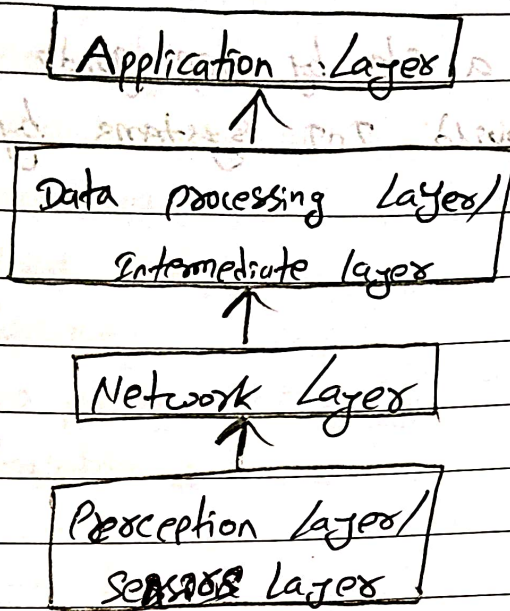
- It helps them to decide:

- i) what to build
- ii) Who will use it
- iii) What data it will collect

### Areas of IOT decision framework

- i) user experience: It describes how users interact with it.
- ii) Data: what data will be collected from the environment, how it will store and how to analyze it.
- iii) Business: It describes which problem will be solved.
- iv) Security: It is used to implement out the data and device will be protected.
- v) operations: How will the system be managed after deployment like device updates, maintenance.
- vi) Devices: It shows which hardware devices we are using like camera, microcontroller, wifi.

## IOT Architecture



→ IOT architecture is the structure or design of how IOT devices (components) work together from sensing the environment to sending data, processing the data and finally showing the result to the user. It is divided into four layers: →

i) Perception Layer

→ It is also known as Sensor Layer which is responsible for sensing the data or collecting the information from the environment.

Devices used: i) Sensors ii) Actuators iii) Identification

i) Sensors: Temperature sensor, Humidity sensors, GPS

ii) Actuators: Motors, LED

iii) Identification: RF-ID, NFC



## ② Network layer

- This layer transfers data from sensing devices to the cloud using internet or communication network.

Technology used:

- i) wifi
- ii) Ethernet
- iii) 4G, 5G

Protocols used to transfer data.

- i) HTTP
- ii) IPV4, IPV6
- iii) TCP / IP

## ③ Data processing layer / intermediate layer / middleware layer.

- This layer stores the data, processes and analyzes the data. It also makes the decision based on the data.

Technology used

- i) Cloud platform: AWS, Azure, Google cloud
- ii) Databases: MongoDB, MySQL
- iii) AI & ML: for analyzing and processing data
- iv) Decision Engine: if, then, etc.

## ④ Application layer:

- This layer displays the information to the user and allows them to control the IoT device.

Examples: mobile applications, web applications

## Microcontroller vs Microprocessor

- i) It is a small computer on a single chip with CPU, input, output ports and memory.
  - It is a chip that contains only CPU, it needs external memory and peripheral devices.
- ii) Designed to control specific device or task.
  - Designed to process data and run complex applications.
- iii) CPU + RAM + ROM + Input-output ports etc.
  - only CPU, RAM, ROM input-output ports are external.
- iv) It is cheap as compared to microprocessor.
  - It is more expensive.
- v) It is small and compact.
  - It is normally larger.
- vi) It requires low power consumption.
  - It requires high power to operate.
- vii) It is generally slower than microprocessor (usually MegaHertz Range).
  - It is much faster (GigaHertz Range).
- viii) Example: Arduino, ESP 32.
  - Example: Intel i5, AMD Ryzen.



## # Arduino UNO:

→ Arduino UNO is a micro controller board based on Atmega328P microcontroller chip. It is designed to make it easy for beginners and professionals to create electronic project and IoT devices by allowing them to connect sensors actuators and electronic components.

### Different pins in Arduino UNO

① Power pins (left side area near the USB port)

→ This hub powers the sockets used to provide power supply to our code (3.3 V pin, 5 V pin, Ground pin (GND), VIN). It is also used to power external battery or adapter.

② Analog pins (located on the left side of the board)

- A0, A1, A2, A3, A4, A5

- It reads analog signal like temperature, humidity, light intensity and convert them into digital numbers using ADC (Analog to digital converter) inside the Atmega328P.

③ Digital Pins (D0 - D13)

- It is located on the right side of the board D0 - D13.  
Function:

- It reads and sense data in two modes.  
Input mode - Read signal (on / off)  
output mode - Sends digital signal